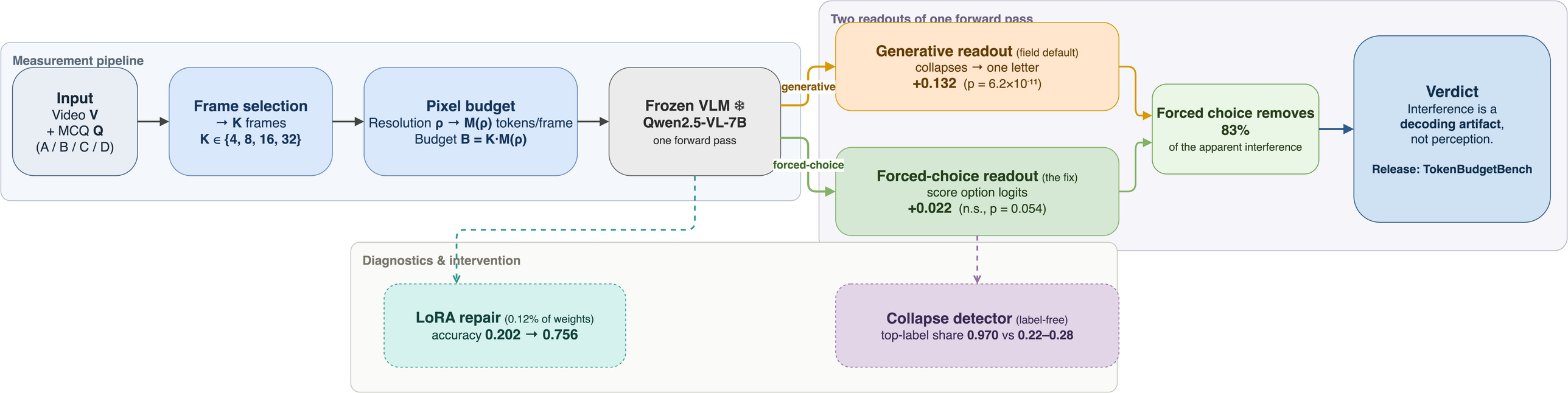

Measurement pipeline
Input
Video V
+ MCQ Q
(A / B / C / D)
Frame selection
→ K frames
K ∈ {4, 8, 16, 32}
Pixel budget
Resolution ρ → M(ρ) tokens/frame
Budget B = K·M(ρ)
Frozen VLM ❄
Qwen2.5-VL-7B
one forward pass
generative
forced-choice
Two readouts of one forward pass
Generative readout (field default)
collapses → one letter
+0.132 (p = 6.2×10⁻¹¹)
Forced-choice readout (the fix)
score option logits
+0.022 (n.s., p = 0.054)
Forced choice removes
83%
of the apparent interference
Verdict
Interference is a
decoding artifact,
not perception.
Release: TokenBudgetBench
Diagnostics & intervention
LoRA repair (0.12% of weights)
accuracy 0.202 → 0.756
Collapse detector (label-free)
top-label share 0.970 vs 0.22–0.28